

1. Please describe the functions of data plane and control plane in each router.
2. Please describe the functions of Time to Live and Upper Layer fields in IPv4 header.
3. What is CIDR?
4. Please describe the idea of route aggregation.
5. Why must routers select the route with a more specific subnet identifier for packet forwarding to ensure correctness?
6. Please describe the operations of link-state routing algorithm.
7. Please use an example to explain why link-state routing algorithm would cause route oscillation.
8. Please describe the operations of distance-vector routing algorithm.
9. Please describe the problem of count-to-infinity for distance-vector routing algorithm.
10. Please show how poisoned reverse solves the problem of count-to-infinity.
11. Please compare link-state and distance-vector algorithms in terms of message complexity, convergence speed and robustness.
12. Please describe the concept and benefits of hierarchical OSPF.
13. Please explain how Internet uses intra-AS and inter-AS routing to achieve scalable routing.
14. Please describe the functions of eBGP and iBGP sessions.
15. Please describe the generation of forwarding-table entries in a router with BGP and OSPF.
16. Please describe how hot potato routing works.
17. Please use examples to examine the operations of policy for BGP.